StyleFinder - AI Clothing Retrieval

Project Overview

StyleFinder is a FastAPI-based web service that enables visual fashion search using a pretrained **ViT-B/32** model from the **transformers** library. Users can upload images or capture photos via the web UI to retrieve similar clothing items from a local gallery.

Architecture

```mermaid
flowchart TD
    subgraph Client
        UI[Web UI (HTML/JS)] -->|HTTP| API[FastAPI Server]
    end
    subgraph Server
        API --> Model[ViT-B/32 Model]
        Model --> Gallery[Image Gallery (2GB)]
    end
    subgraph Deployment
        API -->|tunnel| Cloudflare[Cloudflare Tunnel]
        Cloudflare --> Internet[Public URL]
    end
```

Key Files & Their Roles

File	Purpose
`cir_server.py`	FastAPI application, serves UI and API endpoints, includes new camera UI button.
`Dockerfile`	Container definition for deployment; respects `PORT` env var.
`requirements.txt`	Python dependencies (fastapi, uvicorn, torch, transformers, timm, python-pptx, fpdf).
`prepare_cloud_package.py`	Collects only required assets (model checkpoints, selected images) into `deploy_package/`.
`upload_to_hf.py`	Helper script to create a HuggingFace repo and upload the deployment package.
`start_background_server.vbs`	Windows VBS script that starts the FastAPI server **and** the Cloudflare tunnel, writes the public URL to `tunnel_url.log` and creates a clickable shortcut.
`stop_background_server.bat`	Terminates the server and tunnel processes.
`StyleFinder_Colab.ipynb`	GoogleColab notebook that launches the service in the cloud (free alternative to a 24/7 tunnel).
`StyleFinder_Documentation.md`	This markdown file - source for the PDF documentation.
`generate_docs.py`	Python script that converts the markdown to a PDF (`StyleFinder_Documentation.pdf`) and creates a PowerPoint deck (`StyleFinder_Presentation.pptx`).

Tools & Libraries Used

- **FastAPI** - web framework for the API.
- **Uvicorn** - ASGI server.
- **torch & transformers** - model inference.

- **timm** - vision model utilities.
- **cloudflared** - provides a public HTTPS URL via Cloudflare Tunnel.
- **python-pptx** - programmatic PowerPoint generation.
- **fpdf** - lightweight PDF generation from text.
- **Docker** - containerization for reproducible deployment.
- **GoogleColab** - free compute environment for a 24/7-like demo.

Deployment Options

1. **Local + Cloudflare Tunnel** - Run `start_background_server.vbs`; a public URL is created while the laptop stays on.
2. **Docker Container** - Build with `docker build . -t stylefinder` and run with `docker run -p 8000:8000 stylefinder`.
3. **GoogleColab Notebook** - Execute `StyleFinder_Colab.ipynb` to launch a temporary server with a tunnel.
4. **Future Cloud Hosting** - Upload `deploy_package/` to HuggingFace Spaces (requires a Pro plan) or other cloud providers.

Generating Documentation & Presentation

Running the following command will produce both artifacts:
```bash
python generate_docs.py
```
- `StyleFinder_Documentation.pdf` - compiled PDF version of this markdown.
- `StyleFinder_Presentation.pptx` - a ready-to-present slide deck.

Next Steps

- Consider a paid HuggingFace Space for true 24/7 hosting.
- Replace the static image gallery with a cloud-based storage bucket for scalability.
- Extend the UI to support mobile cameras directly (e.g., via `<input type="file" capture="environment">`).